Multiplication is lambda definable

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openlogic:lambda-calculus/lambda-definability:local-snapshot-20261013; CC BY 4.0

Editorial note. The counted statement is the author proposition labelled prop:mult-ld (482-488); the proof span 490-501 is the author proof environment that immediately follows it.

Editorial note. The editorial note on eta-reduction and the prob about the alternative term Mult' lie outside the counted proof and are excluded.

Editorial scope. One original result of the Open Logic Project unit lambda-calculus/lambda-definability, counted once: lambda definability of multiplication. Only the counted proposition and its own author proof are retained; the source is reproduced without mathematical changes and keeps the CC BY 4.0 licence and the author attribution.

Allowed standard prerequisites (editorial).

- Church numerals as iterators. The Church numeral of m applied to f and x applies f to x exactly m times.. It underwrites both halves of the iteration count in the proof (490-501).
- Lambda definability. A number-theoretic function f is lambda definable when some term applied to the Church numerals of the arguments reduces to the Church numeral of the value.. It is the criterion the displayed Mult term must meet (482-501).

Proposition 0.1. *Multiplication is !!lambda definable by the term*

$$\mathrm{fn}\, Mult\, \mathrm{id}\, \lambda[ab][\lambda[fx][a(bf)x]]$$

Proof. To see how this works, suppose we apply fn $Mult$ to Church numerals $\underline{n}$ and $\underline{m}$: fn $Mult\, \underline{n}\, \underline{m}$ reduces to $\lambda[fx][\underline{n}(\underline{m}\, f)x]$. The term $\underline{m}f$ defines a function which applies f to its argument m times. Consequently, $\underline{n}(\underline{m}f)x$ applies the function "apply f m times" itself n times to x. In other words, we apply f to x, $n \cdot m$ times. But the resulting normal term is just the Church numeral $\underline{nm}$. □

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